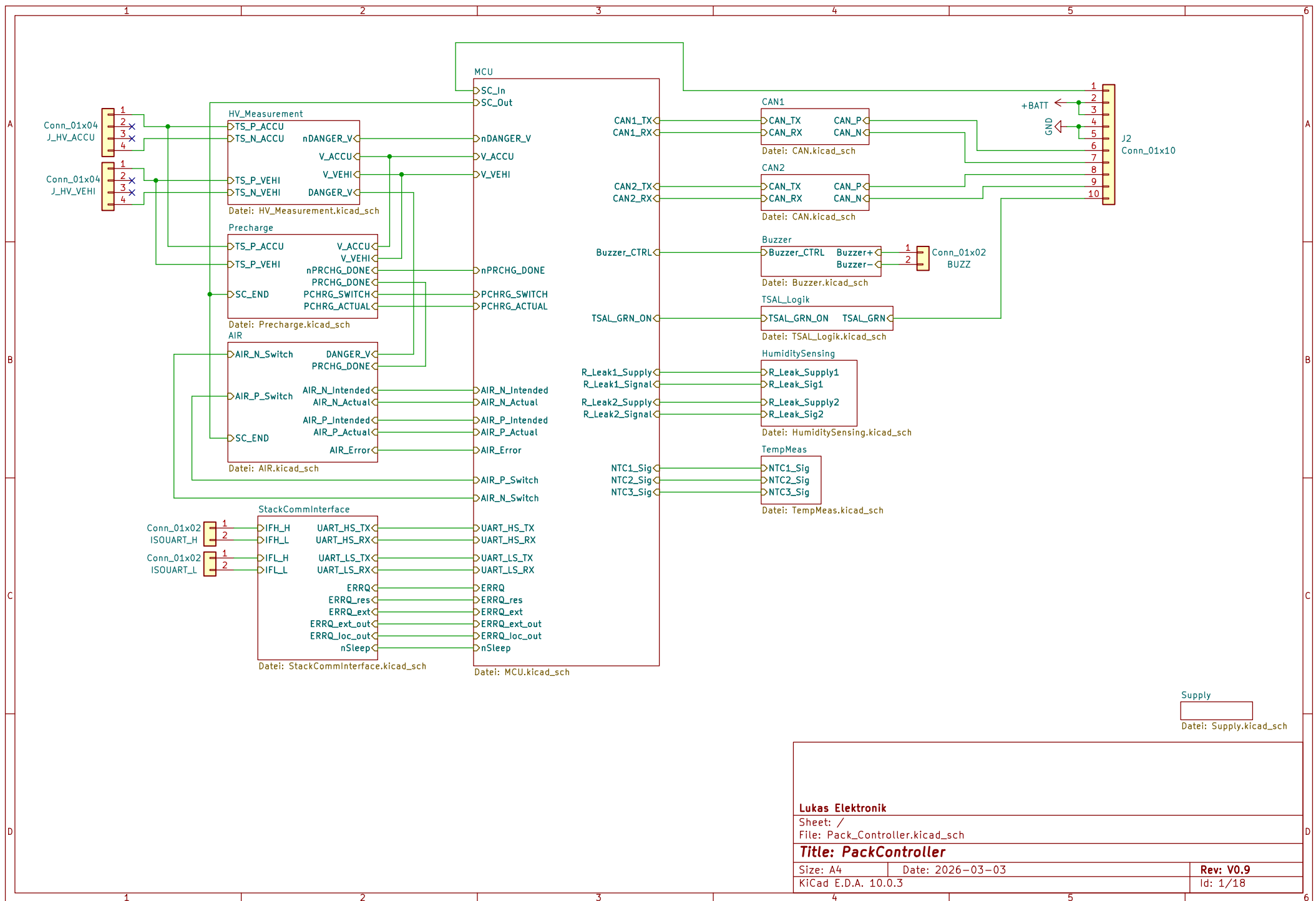
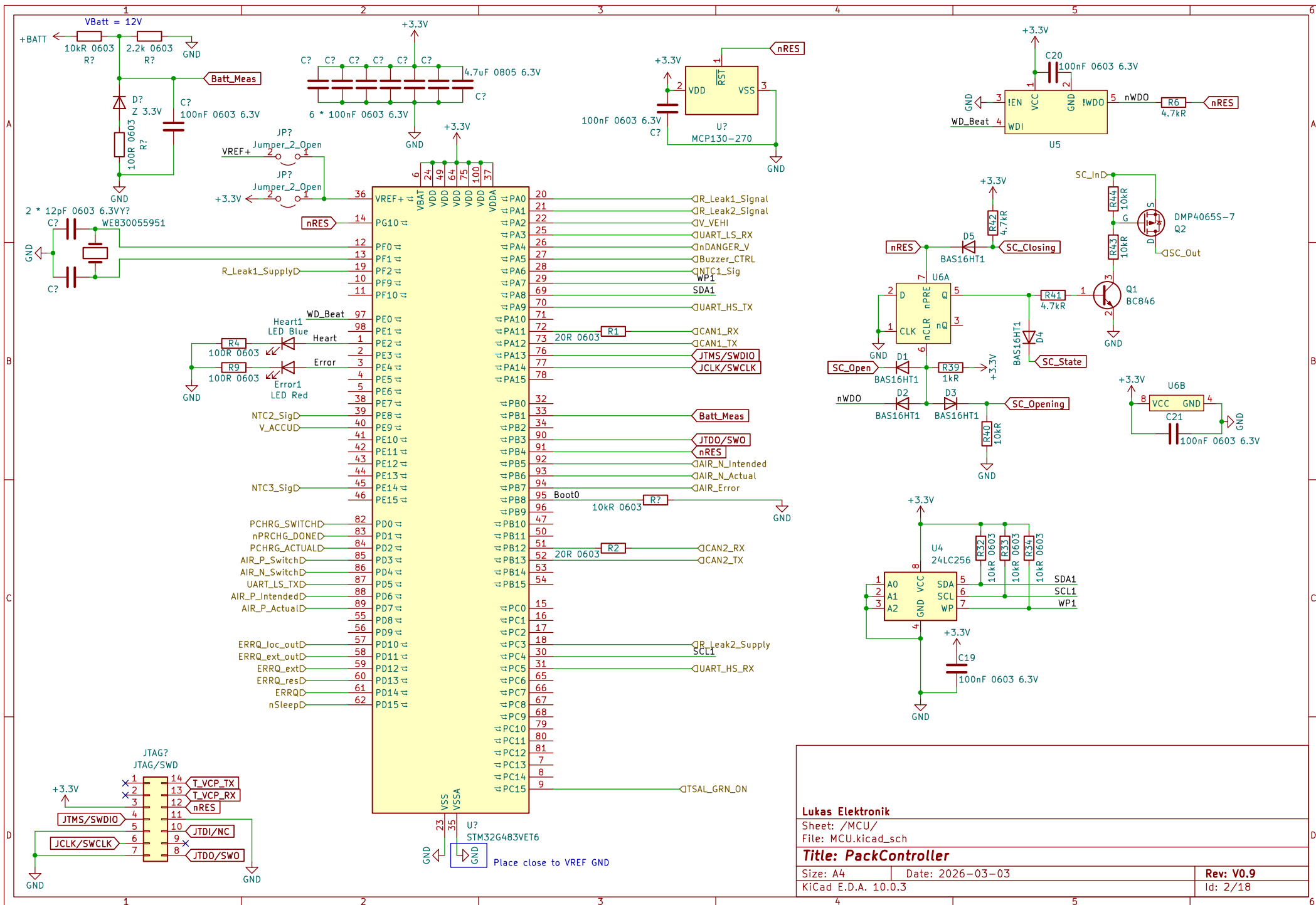
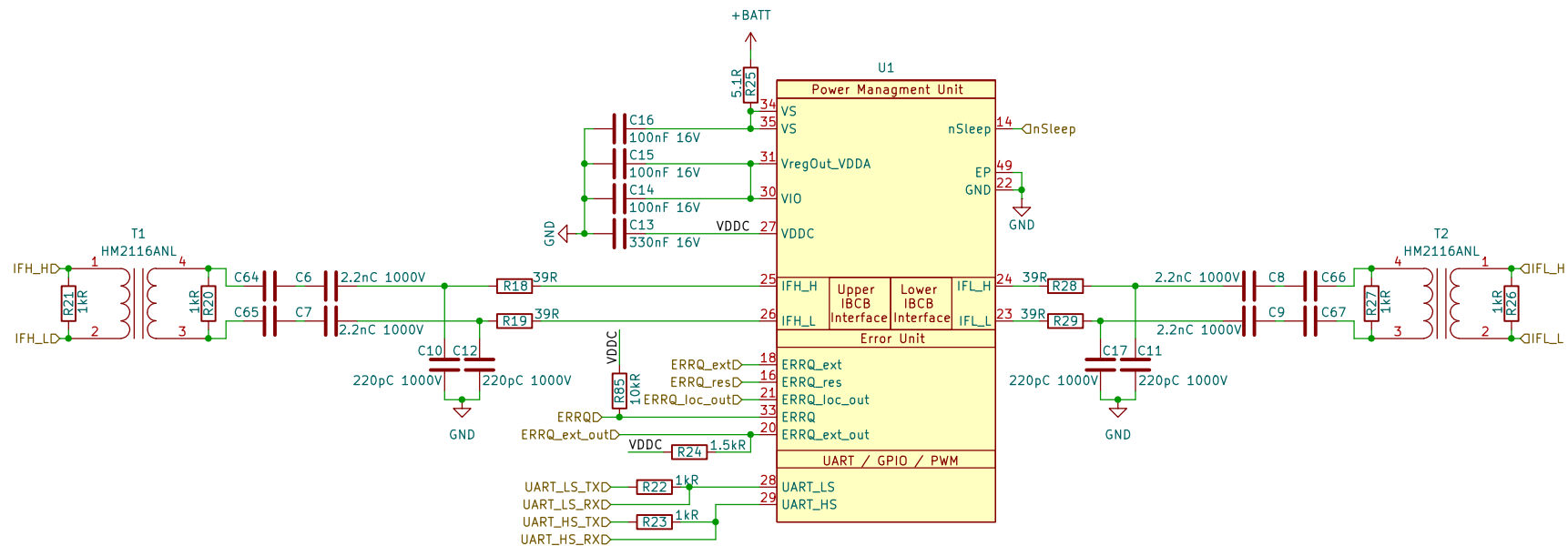








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Sheet: /StackCommInterface/
File: StackCommInterface.kicad_sch

Title: PackController

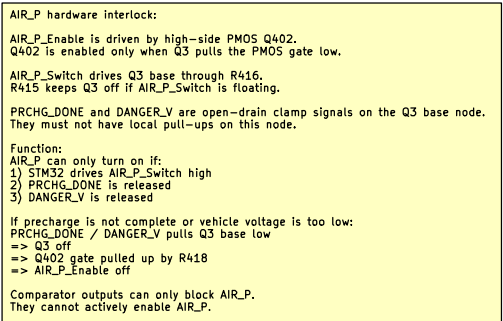
Size: A4

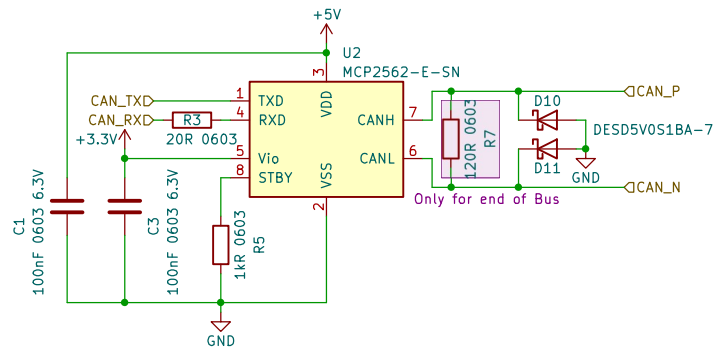
Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 3/18





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Sheet: /CAN1/
File: CAN.kicad_sch

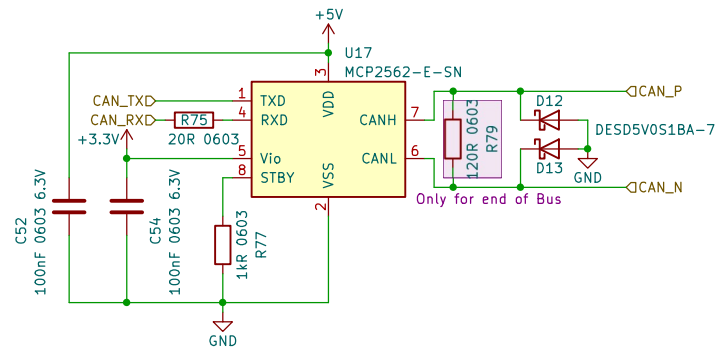
Title: PackController

Size: A4 Date: 2026-03-03

KiCad E.D.A. 10.0.3

Rev: V0.9

Id: 5/18



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Sheet: /CAN2/
File: CAN.kicad_sch

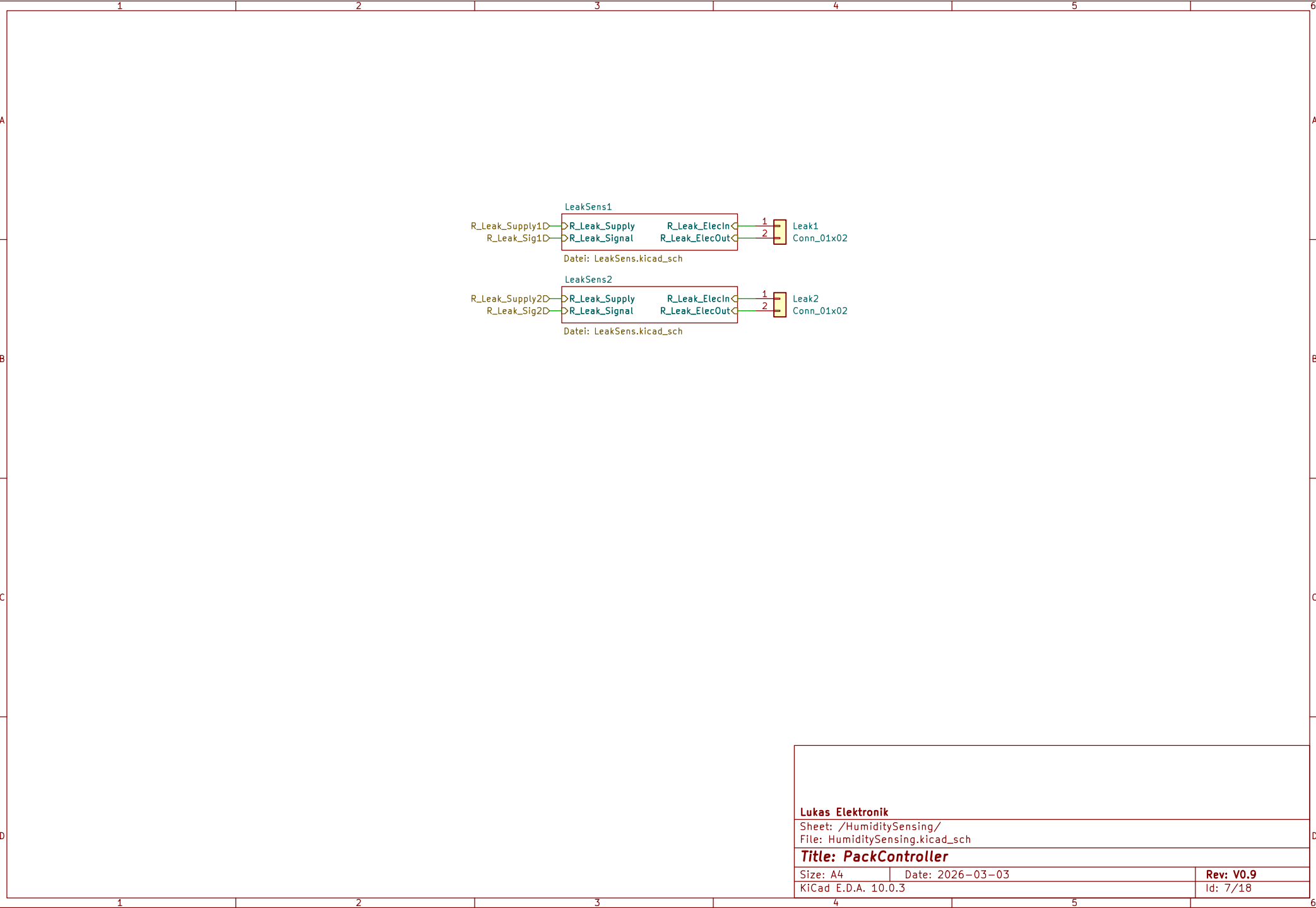
Title: PackController

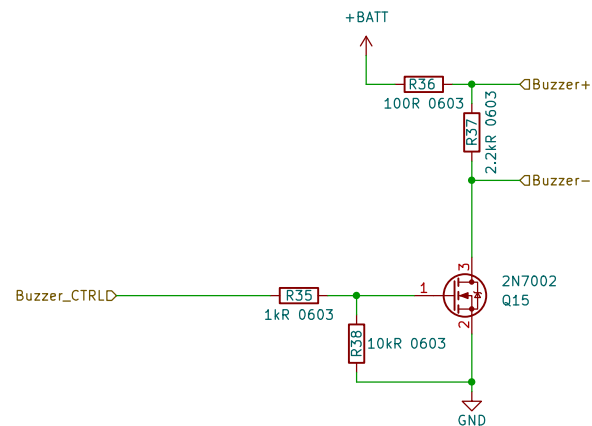
Size: A4 Date: 2026-03-03

KiCad E.D.A. 10.0.3

Rev: V0.9

Id: 6/18





Passive piezo buzzer driver.
 Target part: CPT-2305-90PM, externally driven, 12 Vp-p, 4 kHz, typ. 24 nF.

Drive with STM32 PWM:
 $f_{PWM} \approx 4 \text{ kHz}$
 $duty \approx 50 \%$

Q15 is a low-side NMOS switch.
 R37 discharges the piezo during MOSFET off-time:
 $\tau = R37 * C_{PIEZO} \approx 2.2k * 24nF \approx 53\mu s$.

R36 limits peak current and EMI.
 No additional series coupling capacitor required.

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Sheet: /Buzzer/

File: Buzzer.kicad_sch

Title: PackController

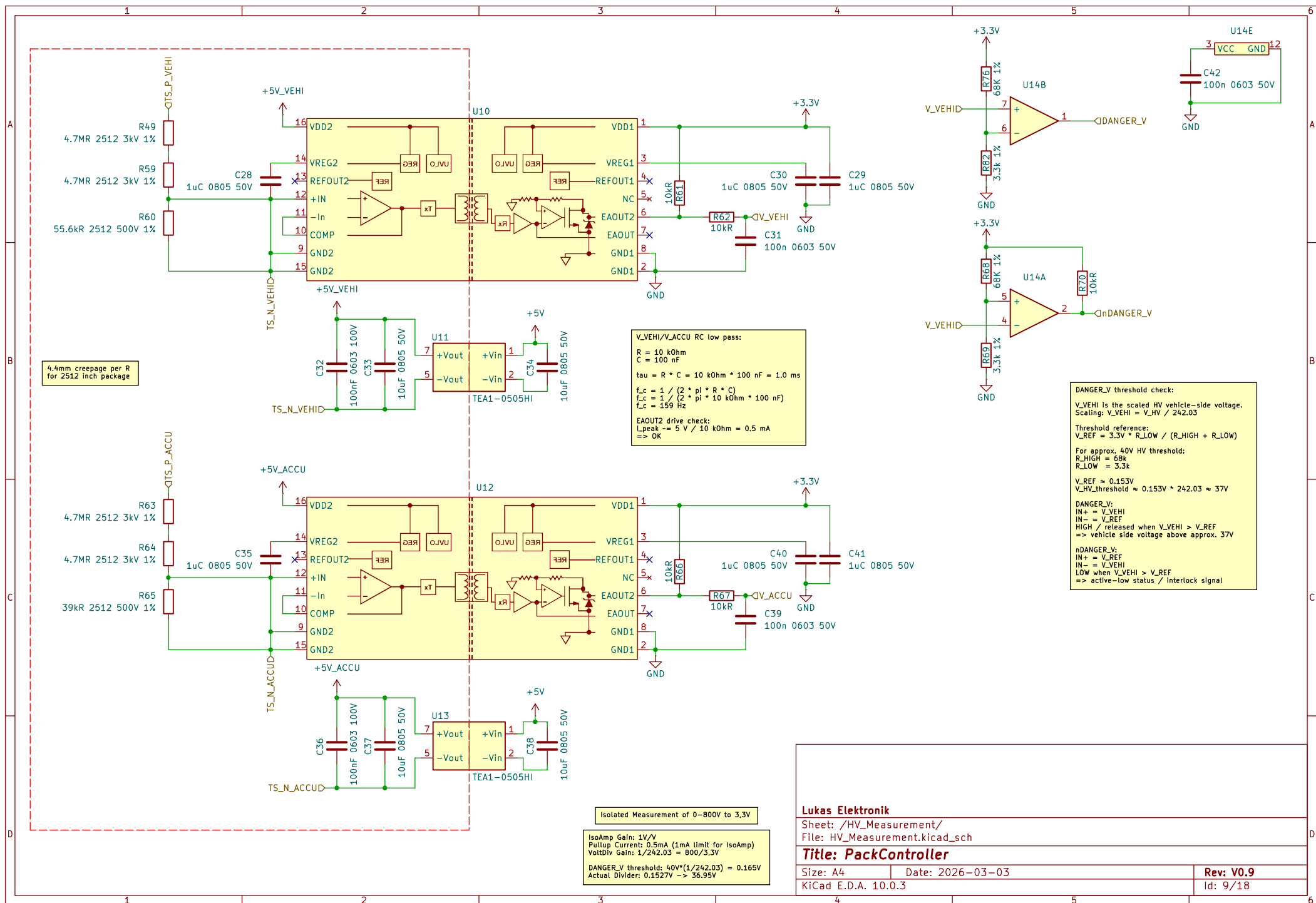
Size: A4

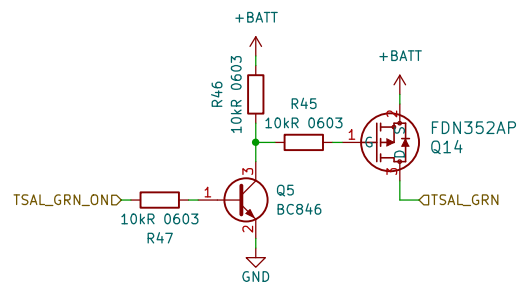
Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 8/18





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Sheet: /TSAL_Logik/

File: TSAL_Logik.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 10/18

Peak Strom bei 800V limitiert durch PTC's R25 = 1kOhm für 2S. -> 0,8A
Toleranz PTC_{EL} = 30% -> 0,8/0,7 = 1,14A
Q7 (IXTA08N100D2HV) Max Cont = 1,8A @ 100°C T_{Die}, Pulsed peak = 5A

Zwei Dioden in Reihe für
Isolation zwischen Fahrzeug
und HV-Batt
3*V_{MAX_HVBatt} -> 3*800 =
2400V
V_{RSM} 4kV
1,7V V_f @100mA

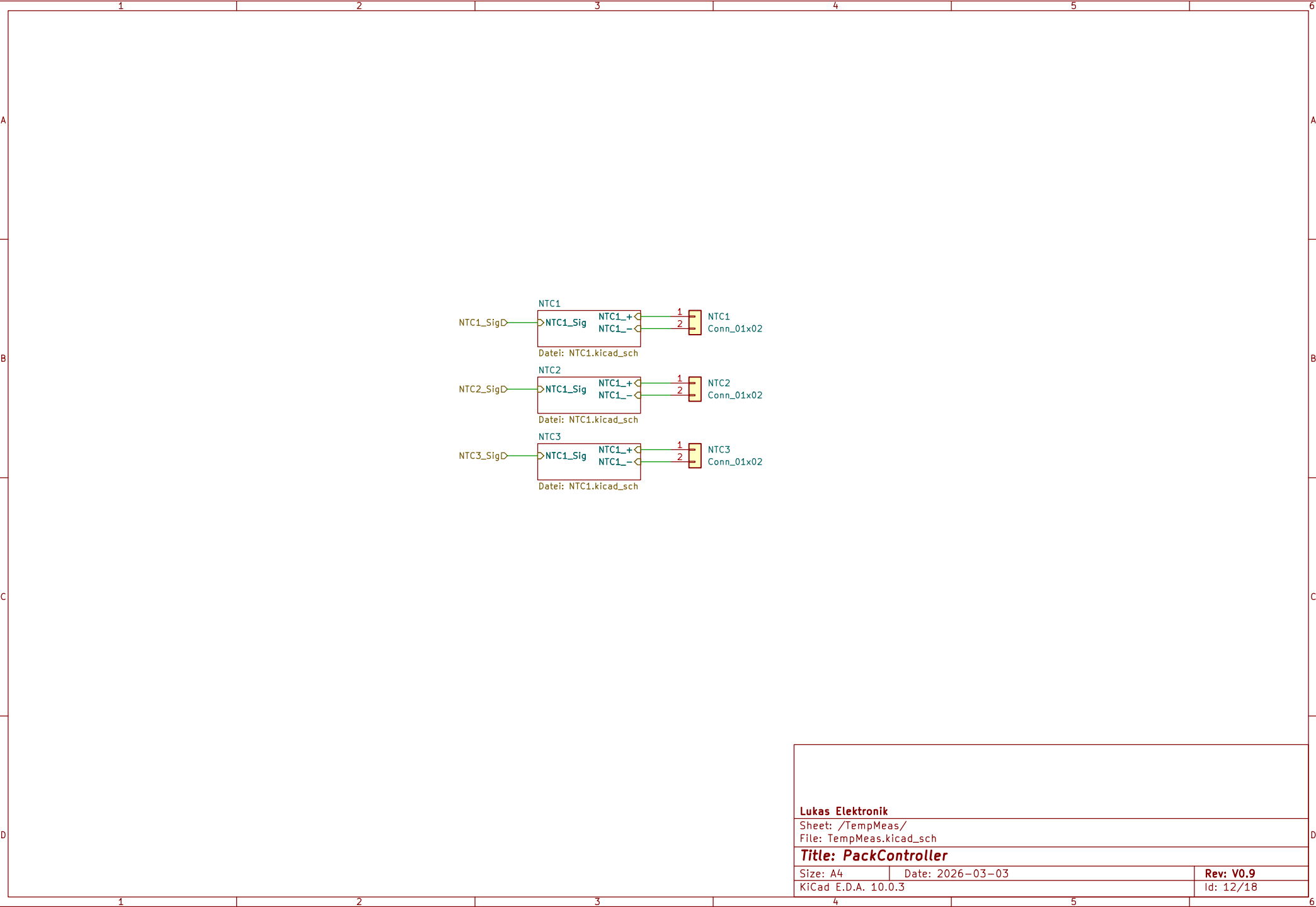
Precharge logic state
PC open: PCHRG_ACT = GND
PC closed: PCHRG_ACT = 3.3V

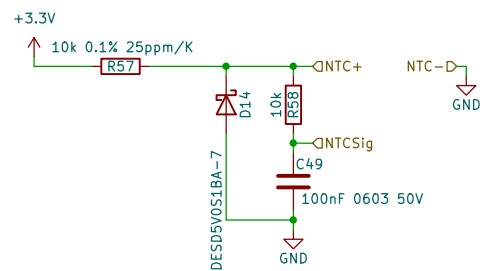
Haltezeit Relais K1
R = 800Ohm
C = 47uF (20%) min = 37,6uF
tau = 3,008ms
Vt = 0,55V (Drop Out KTL12)
Vi = 5V
 $\tau = -R \cdot C \cdot \ln(V_t/V_i)$
tau = -3,008ms * ln(0,55V/5V)
tau = -3,008ms * -2,207
tau = 6,64ms
Q6_t_{off} = 4,8ms

Q6 PRECHARGE GATE TIMING / WORST CASE
Bestückung:
R50 = 10 kOhm
R51 = 10 kOhm
R52 = 1 kOhm
R53 = 10 kOhm
R54 = 1 kOhm
C22 = 200 nF
U7, U8 = TC1T1004
D6 + D7 = ca. 1.7 V
Annahmen:
+5 V liegt vor D6/D7
V_{C,U7} = 5.0 V - 1.7 V = 3.3 V
V_{CEsat,U7} max = 0.3 V
=> V_{G,Q6,final,worst} = 3.3 V - 0.3 V = 3.0 V
U7 Eingang:
IF_{U7,min} = (10 V - 1.6 V) / 1 kOhm = 8.4 mA
IF_{U7,max} = (16 V - 1.25 V) / 1 kOhm = 14.8 mA
IF_{limit} = 60 mA
=> U7 LED Strom OK
R54 Verlustleistung:
P_{R54} -> ((16 V - 1.25 V)² / 1 kOhm) = 0.218 W
=> R54 >= 0.25 W, Derating beachten
U7 Ausgangsreserve:
IC_{U7,min} = CTR_{min} * IF_{U7,min}
IC_{U7,min} = 1.60 * 8.4 mA = 13.4 mA
U7 Strombedarf bei V_G = 3.0 V:
L_{R51} = 3.0 V / 10 kOhm = 0.30 mA
L_{R52,U8} = (3.0 V - 1.6 V) / 1 kOhm = 1.40 mA
L_{C22_start} = 3.0 V / 10 kOhm = 0.30 mA
L_{U7_req,max} = 2.0 mA
=> U7 Reserve: 13.4 mA / 2.0 mA = Faktor 6.7
=> Gate-Endspannung Q6 worst case ca. 3.0 V

Q6 Einschalten:
tau_{ON} = R50 * C22
tau_{ON} = 10 kOhm * 200 nF = 2.0 ms
Q6 Schwelle:
VGS(th) = 1.0...2.5 V
Worst case fuer sicher leitend: VGS(th,max) = 2.5 V
t_{ON} = -tau_{ON} * ln(1 - VGS(th,max) / V_{G,final})
t_{ON} = -2.0 ms * ln(1 - 2.5 V / 3.0 V)
t_{ON} = 3.6 ms
=> Q6 erreicht VGS(th,max) nach ca. 3.6 ms
Q6 Ausschalten:
Konservativ: Entladung nur ueber R50 + R51
R52/U8-Pfad ignoriert, daher maximale OFF-Zeit.
tau_{OFF} = (R50 + R51) * C22
tau_{OFF} = (10 kOhm + 10 kOhm) * 200 nF = 4.0 ms
Worst case Startpegel fuer OFF:
V_{G,start,max} = 3.3 V
Sicher nicht leitend unter VGS(th,min) = 1.0 V:
t_{OFF} = tau_{OFF} * ln(V_{G,start,max} / VGS(th,min))
t_{OFF} = 4.0 ms * ln(3.3 V / 1.0 V)
t_{OFF} = 4.8 ms
=> Q6 sicher unter VGS(th,min) nach ca. 4.8 ms
U8 / PCHRG_ACT:
IF_{U8,min} = (3.0 V - 1.6 V) / 1 kOhm = 1.4 mA
CTR_{U8,min} = 56 % bei IF = 1 mA
IC_{U8,min} = 0.56 * 1.4 mA = 0.78 mA
Fuer PCHRG_ACT -> 3.0 V an R53:
L_{R53} = 3.0 V / 10 kOhm = 0.30 mA
=> U8 Reserve: 0.78 mA / 0.30 mA = Faktor 2.6
=> PCHRG_ACT wird worst case sauber HIGH
Ergebnis:
Q6 ON sicher nach ca. 3.6 ms
Q6 OFF sicher nach ca. 4.8 ms
U7, U8 CTR-Reserve ausreichend

Precharge 90% check:
V_{ACC} and V_{VEH} are equally scaled HV measurements.
V_{ACC_90} = V_{ACC} * R_{LOW} / (R_{HIGH} + R_{LOW})
R_{HIGH} = 1.00M
R_{LOW} = 9.09M
V_{ACC_90} ≈ 0.900 * V_{ACC}
PRCHG_DONE:
IN+ = V_{VEH}
IN- = V_{ACC_90}
HIGH when V_{VEH} >= 0.9 * V_{ACC}
Used as MCU status input.
AIR_P Interlock:
Same comparator condition is also used as open-drain clamp without pullup at AIR_P driver.
Comparator can only block AIR_P, not enable it.
Firmware should close AIR_P only after V_{VEH} >= 95% * V_{ACC} with debounce.
Hardware lockout threshold: 90%





External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

$$R_{NTC} = R_{PULLUP} * x / (1 - x)$$
$$x = ADC_{raw} / ADC_{max}$$

R58/C49 form ADC input filter:
 $\tau = 10k * 100nF = 1ms$, $f_c \approx 159Hz$.

Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

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Sheet: /TempMeas/NTC1/

File: NTC1.kicad_sch

Title: PackController

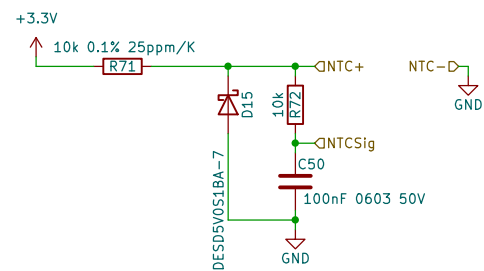
Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 13/18



External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

$$R_{NTC} = R_{PULLUP} * x / (1 - x)$$
$$x = ADC_{raw} / ADC_{max}$$

R58/C49 form ADC input filter:
 $\tau = 10k * 100nF = 1ms$, $f_c \approx 159Hz$.

Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

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Sheet: /TempMeas/NTC2/

File: NTC1.kicad_sch

Title: PackController

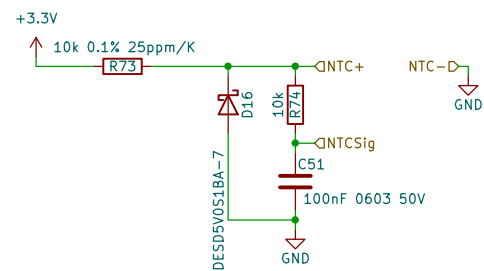
Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 14/18



External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

$$R_{NTC} = R_{PULLUP} * x / (1 - x)$$
$$x = ADC_{raw} / ADC_{max}$$

R58/C49 form ADC input filter:
 $\tau = 10k * 100nF = 1ms$, $f_c \approx 159Hz$.

Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

Lukas Elektronik

Sheet: /TempMeas/NTC3/

File: NTC1.kicad_sch

Title: PackController

Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 15/18

Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor,
parallel to the electrodes, for open-wire detection.
 $R_SENSOR = R_LEAK || R_EOL$

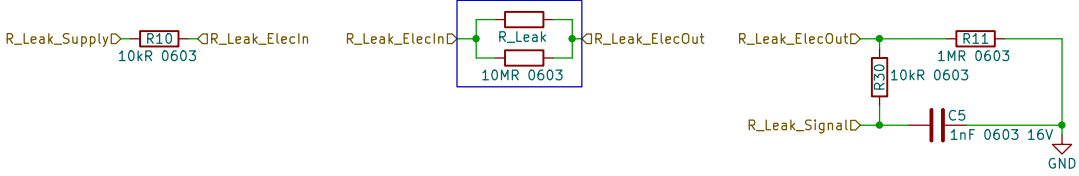
Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/HI-Z.
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$
 $R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY$

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: ADC ≈ 0
dry connected: R_SENSOR ≈ 10M
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: R_SENSOR < 3M
leak: R_SENSOR < 1M
severe leak: R_SENSOR < 300k



Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor,
parallel to the electrodes, for open-wire detection.
 $R_SENSOR = R_LEAK || R_EOL$

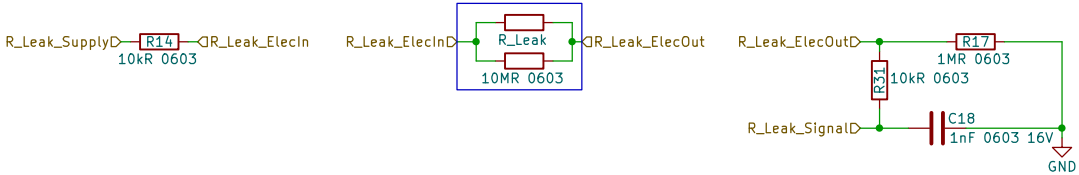
Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/HI-Z.
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$
 $R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY$

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: ADC $\approx$ 0
dry connected: R_SENSOR $\approx$ 10M
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: R_SENSOR < 3M
leak: R_SENSOR < 1M
severe leak: R_SENSOR < 300k



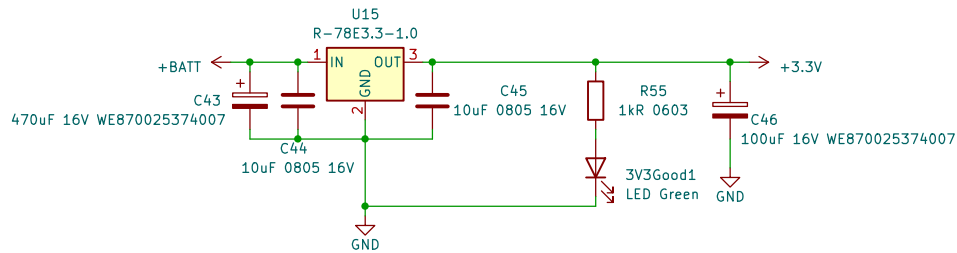
Lukas Elektronik		
Sheet: /HumiditySensing/LeakSens2/		
File: LeakSens.kicad_sch		
Title: PackController		
Size: A4	Date: 2026-03-03	Rev: V0.9
KiCad E.D.A. 10.0.3		Id: 17/18

+3V3 Supply worst-case estimate:

Main loads:
STM32G483 active incl. peripherals: -50 mA
ADuM3190 VDD1, 2x: -4 mA
24LC256 EEPROM write: -3 mA
MCP2562 VIO, 2x worst-case: -1 mA
ADCMP393 comparators + logic: <1 mA
Pullups, dividers, GPIO drivers, LEDs: -20...30 mA

Estimated worst-case load: -80...90 mA
Selected supply: RECOM 3.3 V / 1 A DC/DC

Design margin:
1 A / 90 mA ≈ 11x

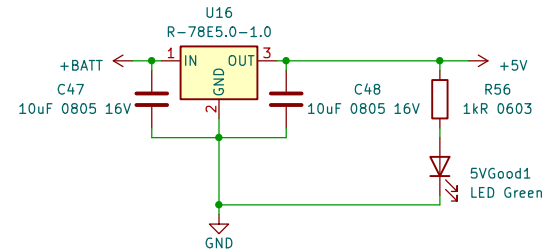


+5V Supply worst-case estimate:

Main loads:
TEA1-0505HI Isolated DC/DC no-load, 3x: -84 mA
TEA1 loads incl. ADuM3190 Isolated sides: -25...30 mA
MCP2562 CAN transceivers, 2x dominant: -140 mA
Misc. reserve / startup / tolerance: -50 mA

Estimated worst-case load: -250...300 mA
Selected supply: RECOM 5 V / 1 A DC/DC

Design margin:
1 A / 300 mA ≈ 3.3x



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Sheet: /Supply/
File: Supply.kicad_sch

Title: PackController

Size: A4
KiCad E.D.A. 10.0.3

Date: 2026-03-03

Rev: V0.9

Id: 18/18